

Geometric Modeling and Experimental Validation of Line Position Estimation Using a Single Infrared Transceiver Sensor

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Abstract

For many years, line following has been one of the most popular approaches for navigating robots due to its simplicity, reliability, and reduced implementation costs. Traditional line followers depend on optical sensor arrays for calculating lateral deviations that occur from the line that the robot follows. Nonetheless, the employment of multiple sensors adds hardware complexities, wiring complications, and effort in signal processing steps. The present paper focuses on a simplistic technique of sensing using only one infrared transceiver sensor. The transmitter and receiver areas of the sensor are modeled as elliptical footprints projected onto a surface. As the sensor sweeps along the guiding line, there are going to be changes in the amount of overlap between the sensing area, the black guiding line, and the background. As a result, it has a varying output for the sensor, and the mathematical model is developed using this relation. The model is validated experimentally using an XY positioning platform that scans the sensor across the guiding line while recording the analog output. The predicted signal agrees well with the measured data, with a root mean square error corresponding to 4.4% of the signal variation range. The effect of sensor mounting height is also examined, and a height of 15 mm is found to provide the most stable signal response. These results indicate that a single infrared transceiver sensor can provide useful position-related information for low-cost line-following mobile robots under the tested conditions.

Keywords

Line-following robot, infrared transceiver sensor, geometric modeling, position estimation, mobile robot navigation

Nomenclature

Symbols	Unit	Description	L_{ws}	mm	Effective sensor workspace length.
a	mm	Semi-major axis of the sensing ellipse.	N		Number of measurement samples.
A_i	mm ²	Intersection area between the sensing region and the reflective line.	N_{in}	sample	Index where the sensor starts entering the line.
A_{max}	mm ²	The maximum effective sensing area of the sensor.	N_{out}	sample	Index where the sensor exits the line.
b	mm	Semi-minor axis of the sensing ellipse.	h	mm	Distance from the sensor lens to the reflective surface.
β	rad	The angle between the Z-axis and the symmetry axis.	θ_e	rad	Divergence angle of the emitter.
d	mm	Distance between the emitter and receiver.	θ_r	rad	Divergence angle of the receiver.
D_R^x	mm	The characteristic projected dimensions of the receiver.	x_c, y_c	mm	Center coordinates of the sensing ellipse.
D_E^x	mm	The characteristic projected dimensions of the emitter.	$y_u(x)$	mm	Upper boundary of the sensing ellipse.
D_R^y	mm	corresponding minor axes	$y_l(x)$	mm	Lower boundary of the sensing ellipse.
D_E^y	mm	corresponding minor axes	Δs	mm/sample	Spatial sampling interval
L	mm	The distance between the centers of the two lenses	S		Sensor output signal.
L_{act}	mm	Active interaction length	S_n		Normalized sensor signal
			S_{min}		Minimum measured sensor signal.
			S_{max}		Maximum measured sensor signal.

S_{exp}		Experimental sensor signal
S_{model}		Predicted sensor signal.
R	mm	Radius of the sensor lens.
$RMSE$		Root mean square error.
w_{line}	mm	Width of the guiding line.

Abbreviations

Abbreviation	Meaning
ADC	Analog-to-Digital Converter
AGV	Automated Guided Vehicle
IR	Infrared
LED	Light Emitting Diode
LDR	Light Dependent Resistor
MR	Mobile Robot
RMSE	Root Mean Square Error

1. Introduction

Line following navigation is considered to be one of the most popular navigation techniques used in various types of mobile robotic systems [1]. This is because of its simplicity, reliability, and low cost of implementation [2]. The technique is normally used in various types of automated guided vehicles, robots used for logistics purposes [3], mobile robots (MR) [4], etc. In most cases, an array of optical sensors is used to detect the lines. The sensors used are normally made up of several emitter-receiver pairs placed on the front side of a robot [5]. Accordingly, the robot is then able to calculate its lateral offset based on the data received from various sensors [6].

Actually, despite the reliable performance of the detection using the various configurations of the sensors, there are various practical issues associated with the implementation of the multi-sensor configuration [7]. Specifically, the main issues associated with multi-sensor configurations are increased system complexity, higher power consumption, and the need for more complex decision-making algorithms to determine the robot's position relative to the guiding line as the number of sensors increases [8].

The infrared reflective sensors are widely used for line-following robots since they are cheap and simple to interface to microcontrollers. Their operation is based on the contrast in reflectivity between the black line and the surrounding surface. In a typical sensor module, an infrared LED illuminates the surface and a photodetector measures the portion of light reflected back to the sensor. Therefore, the output signal varies depending on the local reflectivity conditions under the sensor. Most existing line-following robots use multiple sensors to determine the robot's horizontal position relative to the line. While this configuration is reliable, it increases system complexity and is less suitable for compact or low-cost robot platforms [9], [10].

1.1. Related Work and Research Gap

Previous studies on line-following robots have primarily focused on two research directions. The first

involves sensor configuration, where multiple optical or infrared sensors are arranged in an array to detect the position of the guide path. In these systems, the robot estimates its lateral deviation based on the combined binary or analog feedback of some of the sensor elements [5], [6]. The scheme described is simple to realize and effective for many applications. On the other hand, the accuracy of the system is limited by the number of sensors used and their spacing. Although increasing sensor density improves accuracy, it also requires more inputs, wiring, power, and calibration effort.

The second research area focuses mostly on controller design. Different approaches involving PID controllers, optimal controllers, and intelligent controllers have been proposed for improving tracking precision, response time, and durability [9], [10]. In these studies, the sensing mechanism is treated as one of the available inputs for the system, most often a sensor array. Consequently, there is a lack of focus on the sensing stage, especially when it comes to the problem of whether the infrared transceiver sensor provides adequate information for estimating the linear position.

From a sensing perspective, a single infrared transmitter–receiver pair should be regarded as an area-based detector rather than a simple binary point detector. It means that when the sensor is placed on a reflective surface, the emitted infrared light forms a distributed illuminated area, while the receiver detects the reflected light within its sensitive range [11]. At this point, the sensor response is produced over an area rather than at a single point. During a lateral scan across the guiding line, different portions of this active area cover the black line and the surrounding background. This change in coverage modifies the amount of reflected light received by the detector and produces a measurable change in the output signal. Hence, this solidly indicates that a single infrared transceiver sensor may contain more spatial information than a simple line/no-line output.

Although infrared reflective sensors have been widely used in line-following robots, the geometric relationship between the active sensing region of a single infrared transceiver sensor and the guiding line remains sufficiently quantified for line-position estimation. Existing studies mainly obtain lateral position information by increasing the number of sensors or by improving the control algorithm after the sensing signal has been obtained. Therefore, there remains a research gap in modeling how the spatial sensing footprint of a single infrared transceiver can be used to estimate the position of a guiding line.

1.2. Proposed Approach and Contributions

To resolve this research gap, this study proposes a geometric modeling approach to extract line-position information from the analog response of a single infrared transmitter–receiver sensor. The sensor is modeled as an area-based sensing element rather than a simple point detector. Specifically, the emission and

reception fields are approximated as elliptical footprints on the reflective surface, and their effective overlap is used to describe the active sensing region. As the sensor moves across a black guiding line, the overlap between this active region, the line, and the surrounding background changes continuously, resulting in a corresponding variation in the sensor output. The practical work of this study consists of developing the geometric sensing model, formulating the relationship between the overlap area and the measured signal, and validating the model through controlled experiments. An XY positioning system was used to scan the sensor laterally across a fixed guiding line while recording the analog output signal. The predicted response obtained from the proposed model was compared with the measured data, and the influence of sensor mounting height was also examined. The results are used to assess whether a single infrared transceiver sensor can provide sufficient position-related information for low-cost line-following applications.

The main contributions of this research are summarized as follows:

- A geometric sensing model is developed to describe the effective sensing range of a single infrared transceiver sensor on a reflective surface.
- A line-position estimation method is formulated by relating the variation of the effective overlap area to the measured sensor signal.
- A hardware-simplified sensing strategy is demonstrated, reducing the sensing configuration from a conventional multi-sensor array to a single infrared transceiver sensor.
- The proposed model is experimentally validated using an XY positioning platform, showing close agreement between the predicted and measured sensor signals with low RMSE.

2. Formulation and Method

2.1. Sensor Configuration and Design Parameters

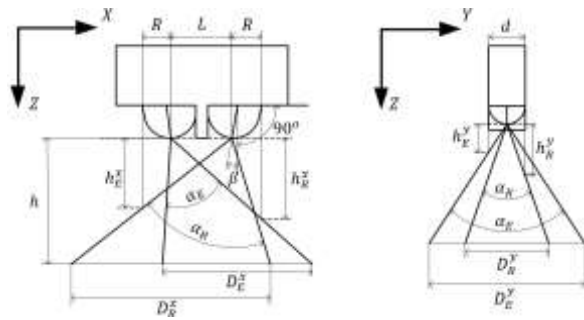


Figure 1: Structure of a transceiver sensor

The design parameters of each line sensor pair are illustrated in Figure 1. In this model, h represents the distance from the LED lens to the reflective surface, R is the radius of the lens, and L is the distance between the centers of the two lenses. The parameter β stands for the angle between the Z-axis and the symmetry axis

of the emitter–receiver pair, defining the relative orientation between the emitter and the receiver.

The parameters α_R and α_E denote the divergence angles of the receiver and emitter, respectively, where the subscript R refers to the receiver and E refers to the emitter. The parameters D_R^x and D_E^x represent the major axes of the elliptical light spots formed on the reflective surface by the received and emitted beams, respectively, while D_R^y and D_E^y represent the corresponding minor axes.

Assumption 1: The distances from both the emitter and receiver lenses to the reflective surface are identical.

Assumption 2: The emitter and receiver lenses have the same diameter.

Assumption 3: The emitter and receiver are symmetrically oriented toward each other with identical angular orientations.

2.2. Geometric Modeling and Position Estimation Method

• Geometric Modeling

This section develops a geometric model describing the relationship between the sensing region of the infrared transceiver sensor and the guiding line. The model is used to establish a quantitative relationship between the sensor output signal and the relative position between the sensor and the line.

When the infrared emitter projects light onto the reflective surface, the emitted beam forms a spatial illumination region. Similarly, the receiver detects reflected light within its receiving range. Due to the directional radiation characteristics of the emitter and the angular sensitivity of the receiver, the emission and reception regions projected onto the reflective surface can be approximated by elliptical distributions.

The effective sensing region of a transceiver sensor corresponds to the overlapping region between the emission and reception distributions on the reflective surface. This region is called the sensor's workspace. Only within this workspace can emitted light be reflected by the surface and detected by the receiver. Therefore, the workspace defines the effective sensing region and contributes to the sensor's output signal.

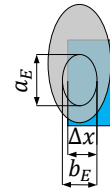


Figure 2. Relative position between the sensor and the line on the reflective surface.

Figure 2 illustrates the relative position between the sensing area and the guide path on the reflective surface. As the sensor moves across the guide path, the overlap between the sensing area and the reflective surface changes. This change in the overlap alters the intensity of the reflected light received by the sensor.

For modeling purposes, the projected sensing region is approximated as an ellipse on the surface. The ellipse is described by:

$$\frac{(x-x_c)^2}{a^2} + \frac{(y-y_c)^2}{b^2} = 1, \quad (1)$$

where a and b signify the semi-major and semi-minor axes of the ellipse, respectively, and (x_c, y_c) represent the coordinates of the ellipse center.

The dimensions of the projected emission and reception regions depend on the sensing height h and the divergence angles of the emitter and receiver beams. Based on geometric projection, the major axes of the emitted and received elliptical light spots can be approximated as

$$D_E^x = 2htan(\alpha_E), \quad (2)$$

$$D_R^x = 2htan(\alpha_R), \quad (3)$$

where D_E and D_R designate the characteristic projected dimensions of the emitter and receiver regions, respectively. These expressions show that increasing the sensor height enlarges the projected sensing region, which later explains the effect of mounting height discussed in Section 3.

- *Overlap Area Model*

The effective sensing region of the transceiver sensor corresponds to the overlapping region between the emission and reception distributions on the reflective surface. This region defines the sensor's workspace.

When the sensing ellipse intersects the reflective guiding line, only the portion of the sensing region that overlaps the reflective surface contributes to the reflected signal received by the sensor.

Let A_i refer to the intersection area between the sensing region and the reflective line. Assuming that the intersection occurs between two points x_1 and x_2 , the intersection area can be expressed as

$$A_i = \int_{x_1}^{x_2} (y_u(x) - y_l(x)) dx, \quad (4)$$

where $y_u(x)$ and $y_l(x)$ represent the upper and lower boundary functions of the overlapping region, respectively.

The maximum effective sensing area of the sensor, denoted as A_{max} , corresponds to the workspace area when the sensing region lies completely over the reflective surface.

- *Relationship Between Reflective Area and Sensor Signal*

The sensor output signal depends on the intensity of the reflected light received by the detector. Since the reflected light intensity is proportional to the reflective area within the sensing region, the sensor signal can be approximated as proportional to the intersection area.

$$S = kA_i, \quad (5)$$

where S represents the sensor output signal, A_i denotes the intersection area between the sensing region and the reflective surface, and k is a proportionality constant related to the sensor's optical characteristics.

To eliminate the influence of the analog-to-digital converter (ADC) resolution, the sensor signal is normalized as

$$S_n = \frac{S_m - S_{min}}{S_{max} - S_{min}}, \quad (6)$$

where

S_m : measured sensor signal

S_{min} : minimum measured signal

S_{max} : maximum measured signal

S_n : normalized sensor signal

When the sensing region lies entirely over the reflective surface, the reflected signal reaches its maximum value and $S_n = 1$. Conversely, when the sensing region lies completely over the black guiding line, the reflected signal becomes minimal and $S_n = 0$. Hence, Eqs. (4)-(6) define the direct relationship between the sensor's geometry and its measured electrical response.

2.3. Position Estimation and Line-Width Reconstruction

Before presenting the experimental results, the position-estimation rule is defined from the proposed geometric sensing model. After signal conditioning and boundary detection, two characteristic sample indices are extracted from the processed signal: N_{in} , denoting the instant at which the sensor active region starts to overlap the black line, and N_{out} , denoting the instant at which the sensor's active region completely leaves it.

For a transverse scan performed with a spatial sampling interval Δs , the total active interaction length is defined as

$$L_{act} = (N_{out} - N_{in}) \Delta s, \quad (7)$$

Based on the geometric interpretation of the sensor workspace, the detected interaction span extends beyond the black-line width alone. Instead, it contains the actual black-line width together with the effective sensor workspace L_{ws} . The reconstructed line width is therefore defined as

$$w_{line} = L_{act} - L_{ws}. \quad (8)$$

This formulation is fundamental to the proposed method because it establishes a direct relationship between the experimentally detected signal boundaries and the physical sensing geometry. Once N_{in} and N_{out} are identified, the line position may also be inferred from the midpoint of the detected interaction interval. Additionally, before estimating the position, the reflective sensor's analog output must be conditioned to reduce noise while preserving the transition from the white background to the black line. In other words, the analog output of the reflective sensor is subject to various types of interference, such as changes in reflection levels, electronic noise, and mechanical interference caused by the motion. When the analog

output is directly processed, it may interfere with the transition from the black line to the white background. To reduce the interference, the procedure was implemented in three stages. First, the analog output was scaled based on the given bounds to make the transition from the black line to the white background more prominent. Second, the scaled analog output was smoothed to reduce random interference while retaining the transition's trend. Finally, the smoothed analog output was processed to identify the two instances of interest: the point at which the active region of the sensor first intersects the black line, and the point at which the active region completely exits the black line. For clarity, the signal-conditioning and boundary-detection procedure described above is summarized in Algorithm 1.

Algorithm 1. Signal conditioning and boundary detection

- Step 1:** Acquire the raw analog signal S_{raw} from the reflective sensor during the scanning process.
- Step 2:** Normalize the signal using the calibration bounds in Eq. (6).
- Step 3:** Apply a Kalman filter to the normalized signal to suppress random noise while preserving the dominant transition behavior.
- Step 4:** Analyze the filtered signal to identify the first transition point where the sensing region begins to intersect with the black line. Record this index as N_{in} .
- Step 5:** Continue analyzing the signal to detect the second transition point where the sensing region completely exits the black line. Record this index as N_{out} .
- Step 6:** Compute the interaction length following Eq. (7).
- Step 7:** Use the detected indices and the geometric relations to reconstruct the guiding-line width.

3. Result and Discussion

3.1. Experimental Setup

The experiments were conducted using a specially designed XY positioning table that can accurately control the relative movement between the reflective sensor and the black guiding line. The positioning table has two orthogonal axes that can move linearly in the X and Y directions using stepper motors and timing belt drives. The system also has micro-stepping drives for higher resolution and limit switches for homing and end-of-travel protection. The system automatically homes before each test run in order to provide a reference for the X and Y axes. The reflective sensor was then moved to the specified test location, and the calibration test was conducted in order to determine the minimum and maximum analog output for the black line and the white background. The reflective sensor module was mounted on the moving platform of the XY table, while the test surface with the black line was fixed at the base platform of the table. This configuration allows for the transverse scanning of the reflective sensor over the black line while maintaining

the relative geometry between the active area of the reflective sensor and the black line, as shown in Figure 3.

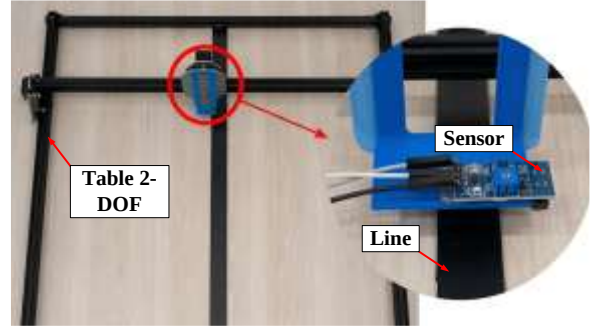


Figure 3. Experimental validation setup: XY positioning table for sensor scanning and a close-up of the reflective sensor above the black guideline.

3.2. Signal Conditioning and Edge Detection

For the validation of the sensor in this case, the sensor was moved across the line over a total distance of 50 mm using 50 sampling points, which is equivalent to a sampling interval $\Delta s = 1 \text{ mm/sample}$. At each sampling point, the analog signal was acquired and processed to detect the sample index N_{in} where the sensor first made contact with the line and the sample index N_{out} where the sensor had fully exited the line, and these indices were used to validate the suggested geometric model.

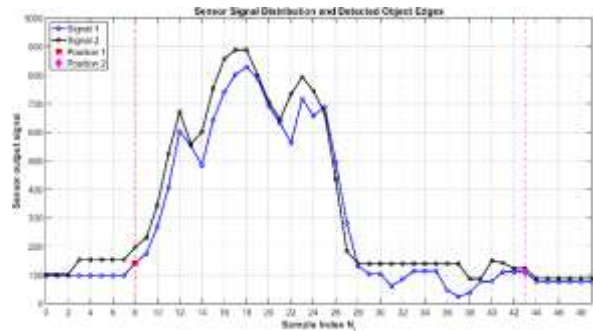


Figure 4. Sensor signal distribution with detected entry and exit boundaries, used to reconstruct the black-line width based on the proposed geometric sensing model.

As seen in Figure 4, the distribution of the sensor signal, as well as the detected boundaries of entry and exit, is also provided. In the validation example, the sensor starts to interact with the black line when $N_{in} = 8$. The sensor leaves the black line when $N_{out} = 43$. The indices of entry and exit define the range of interaction between the sensor and the black line, which is used to validate the geometry.

The sensor workspace was determined to be $L_{ws} = 6 \text{ mm}$. The spatial interval between consecutive samples was determined to be $\Delta s = 1 \text{ mm/sample}$. From the definition provided prior to Section 3, it is clear that $L_{act} = 35 \text{ mm}$. The width of the black line was reconstructed to be $w_{line} = 29 \text{ mm}$. The actual width of the black line used during the experiment was 28 mm.

This outcome gives direct experimental verification of the suggested model of the geometric sensor. More specifically, the distance between N_{in} and N_{out} is consistent with the view that the observed sensor span includes the width of the black line and the sensor's workspace. The strength of the outcome is that it is obtained with a relatively low sampling resolution of 1 mm/sample. This outcome further verifies the proposed sensor model's reliability and suggests that reducing error by increasing the sampling density is possible.

3.3. Validation of the Geometric Model Using Sensor Signal Measurements

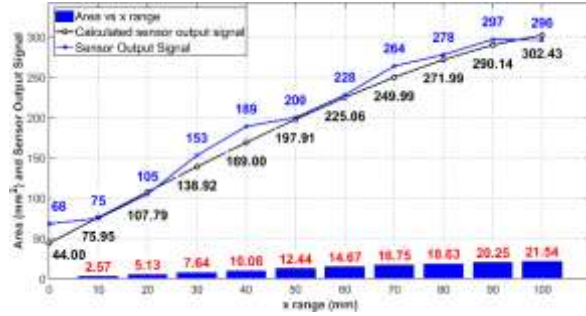


Figure 5. Comparison between the predicted sensor signal from the geometric model and experimental measurements.

Figure 5 illustrates the comparison between the calculated sensor output signal obtained using the intersection area model and the experimental results of the sensor signal. The results obtained demonstrate how the sensor signal is correlated with the predicted trend of the calculated sensing area. It is observed that when the sensor position is increased by 0 mm to 50 mm, the calculated signal is increased from 44.00 to 197.97, while the sensor signal is increased from 68 to 200. Further, at a sensor position of 70 mm, the calculated signal is 250.02, while the sensor signal is 264. This indicates an error of 5.3%. In another instance, at a sensor position of 100 mm, the calculated signal is 302.51, while the sensor signal is 296, indicating an error of 2.2%. Most data points show good agreement between the model and the experiment, with relatively small deviations. Larger discrepancies appear near the low-signal region and transition region, which may be caused by nonuniform reflection and the simplified elliptical approximation.

These minor variations between the theoretical and experimental values may be due to sensor noise, variations in the reflectivity of the surface, and geometric modeling assumptions in the sensing region. In other words, to find out the level of agreement of the theoretical and experimental sensor signal, the root mean square error (RMSE) between the theoretical and experimental sensor signals is computed. A lower RMSE indicates a better agreement between the theoretical model and experimental measurements.

The RMSE is defined as

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i^{measured} - y_i^{model})^2}, \quad (9)$$

where $y_i^{measured}$ represents the measured sensor signal, y_i^{model} represents the signal predicted by the geometric model, and N is the total number of measurement samples.

As a result, RMSE approximates 4.4% of the signal variation range. This demonstrates that the geometric model is a good approximation for sensor response, successfully capturing the relationship between the reflectance area and the signal.

3.4. Effect of Sensor Height on Signal Response

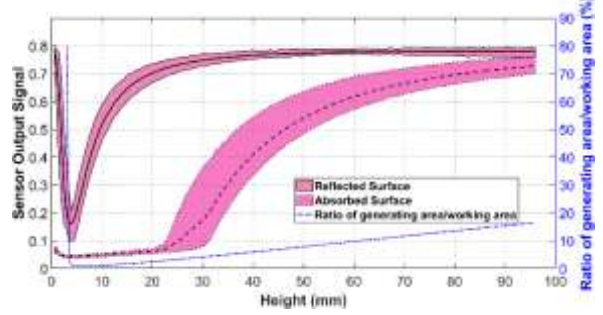


Figure 6. Effect of sensor height on the output signal response.

As shown in Figure 6, the strong effect of the mounting height on the sensor's performance and sensing geometry is clearly shown. Very low heights, especially below 5 mm, cause rapid variations in the output voltage due to the fact that the sensor's sensitivity to any change in the distance between the sensor and the surface is high. With an increase in height, the signal obtained by the reflected surface increases sharply to approach saturation. The signal obtained by the absorbed surface, on the other hand, is initially low and starts increasing when the height reaches higher values, particularly greater than about 25–30 mm. This effect helps in decreasing the difference in terms of reflectivity and absorption at greater mounting heights. Additionally, the ratio of generating area and working area increases with increasing height. This indicates that a larger footprint is generated from the projection at greater distances. Thus, a suitable mounting height is required to achieve a clear signal distinction without causing the sensing area to be too small or too large. Therefore, a mounting height of 15 mm was chosen to satisfy these requirements.

4. Conclusion

The experimental findings show that the suggested sensing model provides a geometric interpretation of the sensor's reflective response. The interaction length is determined by the entry and exit of the response, and the linewidth can then be calculated after accounting for the sensor's workspace. In the validation case, $N_{in} = 8$ and $N_{out} = 43$ yielded $L_{act} = 35\text{mm}$; with $L_{ws} = 6\text{mm}$, the reconstructed line width was 29mm, closely matching the true width of 28mm despite the coarse sampling interval of 1 mm/sample. Moreover, experimental tests on sensor mounting heights have

shown that about 15mm ensures suitable, stable sensing performance and smooth signal variations.

While there is close alignment between the suggested geometric sensing model and experimental findings, there still exist some problems in the current model. First, the emission and reception area of the infrared sensors is assumed to be ideal and ellipsoid-shaped, while the optical path might be distorted by factors such as the shape of the lens and varying intensities of light. Furthermore, the current model is based on the assumption of constant surface reflectivity, although surface reflectivity in practical applications can vary due to material properties, contamination, surface wear, and changes in ambient light. Finally, as experimental validation was performed only on a stationary XY stage, additional verification in mobile robotic systems is required.

The future work will involve incorporating the proposed model for the sensor into the actual control system for a mobile robot and making comparisons of its performance with existing models using multiple sensors.

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